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Markerless joint angle estimation using MediaPipe with a rapid setup for joint moment calculation

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Abstract
This study explores the use of MediaPipe for markerless joint angle estimation and its application in calculating joint moments during sit-to-stand movements. The problem is the absence of an accurate, affordable, and easy alternative to costly, complex marker-based systems that restrict natural movement. Our approach leverages a rapid setup using a single camera to capture motion, making it a cost-effective and accessible solution for motion analysis. We conducted experiments with 15 healthy volunteers, recording their STS movements using a smartphone camera. The videos were processed using MediaPipe to estimate joint angles at the ankle, knee, and hip. These angles were then used to compute angular velocities, accelerations, and joint moments through an inverse dynamics model. We compared the performance of our markerless approach with conventional marker-based systems and MATLAB simulations. The results showed strong consistency in hip and knee joint moment estimations, with correlation coefficients of 0.94 and 0.95, respectively. While the ankle joint demonstrated moderate agreement ($R=0.11$), the findings confirm the method's reliability for lower limb analysis. Our results also highlight the potential of our method for real-time applications in clinical and sports settings. The study underscores the advantages of using a markerless system, including ease of setup, reduced costs, and the elimination of movement restrictions associated with traditional marker-based systems. This research contributes to the field by providing a detailed analysis of STS movements using an innovative, deep learning-based approach. The findings support MediaPipe's effectiveness for biomechanical analysis and its potential in clinical and rehabilitation settings.

Keywords Pose estimation · Deep learning · Motion capture · Biomechanics · Sit to stand

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1 Introduction

To live independently, a healthy person must possess mobility that allows them to perform general daily activities [1]. As people age, their mobility decreases [2], leading to a reduction in joint range of motion and muscle strength, as well as coordination problems. Inability to perform these activities can negatively affect a person's quality of life due to the resulting physical, social, and emotional problems [3]. These issues can arise not only due to aging but also because of musculoskeletal disorders [4] and neurological problems [5]– [6]. Monitoring these conditions is crucial in preventing potential permanent damage if an existing problem is present [7].

Activities such as walking, climbing stairs, carrying objects, and standing up are just a few of the general daily activities [8]. Among these, the action of sitting and standing up is particularly important as it increases the movement range of the upper extremity and trunk. It must allow movement in different directions and have a high load-carrying capacity to support the upper body. Additionally, ensuring that joint movements occur at the correct angles and forces reduces the load on the joints and provides proper biomechanics [9]. Monitoring these movements to ensure they occur at the correct angles and moments is crucial for preventing potential joint damage and diagnosing existing conditions [10].

To better understand the current state of motion analysis technologies, the following literature review is structured according to the basic methodological approaches used in human motion capture systems. The studies are examined in four main groups. Traditional marker-based systems, inertial measurement unit (IMU)-based systems, depth-sensing camera systems, and deep learning-based markerless systems. This classification allows for a clearer comparison of the advantages and limitations of each method.

Various methods are used to track and analyze these movements [11]. In motion analysis applications, kinematic data collection systems are typically used. In these systems, active or passive markers placed on the subject are detected and digitized by cameras positioned in specific locations, and the data is recorded. Kinetic data collection systems also measure ground reaction forces, providing more straightforward and precise data. Both types of data are collected simultaneously by computer systems, filtered, and combined by analysis programs to make them meaningful. The motion capture phase is divided into external and wearable systems. External systems involve sensors placed independently of the subject and can be implemented using RGB cameras or depth sensors. On the other hand, wearable sensor systems can precisely detect muscle, bone, and joint movements. Additionally, precise inertial measurement sensors worn on the limbs can determine the positions of bones [12].

Human motion capture systems can vary in accuracy across different environments and conditions, directly affected by the performance of the systems used [13]. Specifically, systems used for calculating joint moments become more complex due to the data obtained through sensors and cameras and the simulation software used for mathematical calculations. These systems have drawbacks, such as requiring special equipment, having sensors that restrict movement, and needing experts for analysis [14]. To address one or more of these issues, various comparative studies in the literature involve motion capture systems integrated with movement modeling and software applications that can calculate motion parameters similarly to data obtained from camera systems and sensors. These studies develop motion capture systems and compare them with traditional camera-based methods. One such system, Azure Kinect, was compared with a 12-camera Vicon system by Thomas

et al. [15]. The study compared the kinematic and spatiotemporal variables of the sit-to-stand (STS) cycle. It found a high correlation between the two systems overall, although the correlation was low for discrete variables and the medial-lateral pelvic sway range. Additionally, Azure Kinect is deemed potentially useful for clinical assessment and decision-making processes. In a similar study by Moran et al. [16], lower extremity joint angles, segment angles, and spatiotemporal metrics of 21 experienced runners were evaluated using a markerless 8-camera motion capture system and an instrumented pressure treadmill. The results indicated that the system could reliably measure biomechanical joint and segment angles and spatial measurements, with good reliability for discrete kinematic measurements across all planes of motion. This suggests that using a markerless motion capture system can save significant time during data collection while allowing clinicians and researchers to assess changes in working biomechanics.

Studies are being conducted to integrate motion capture applications with camera systems. Particularly in complex motion analysis, markerless images make the evaluation of kinematic data difficult. In this context, Fistch et al. [17] developed an approach called "orientation keypoints" to accurately determine skeletal joint positions and rotation from single-frame RGB images. They noted that using virtual markers to estimate rotation, which is not observable with traditional methods, improved the error rate in joint angles by 48% and provided 93% accuracy in bone rotation.

One of the systems used in motion capture is optical motion capture systems. Shuai et al. [18] compared the OptiTrack system with the IMU-based Perception Neuron for joint kinematics, such as joint range of motion, during functional movements. The results showed that an IMU-based motion analysis system could be effectively applied to clinical functional movement analysis. Another comparative study by Al Borno et al. [19] evaluated the ability of volunteers to measure lower extremity joint kinematics using optical sensors and IMUs in a natural environment. They found that the developed IMU-based system produced results consistent with the optical motion capture system. Another IMU-based study by Slade et al. [20] introduced the OpenSenseRT system, a wearable, open-source system that predicts real-time upper and lower extremity kinematics using inertial measurement units and a portable microcontroller. The study validated OpenSenseRT by comparing it with optical motion capture, demonstrating that it is a low-cost, reproducible system suitable for a wide range of applications. Although IMU-based systems are mobile, sensor drift and the need for physical attachment remain major challenges.

Developments in motion capture systems are increasing, focusing on systems where artificial intelligence plays a more active role. One such study by Bin Housain et al. [21] compared a CNN algorithm with an IMU sensor for motion analysis data. They proposed a deep learning-based approach to estimate joint kinematics in different walking conditions using IMU sensors mounted on roller shoes. The DeepBBWAE-Net framework, combining techniques such as CNN and GRU, showed significant improvements in kinematic predictions. This system represents a major advancement in practical and accessible motion analysis with few IMU sensors, holding promise for biomechanics and rehabilitation research.

In another study, Moro et al. [22] proposed an RGB camera-based, markerless deep learning system for 3D gait analysis. They compared the motion parameters obtained for knee and ankle joint angles of 16 volunteers with a marker-based system. The study highlighted the feasibility of adopting markerless techniques to enhance the accessibility and efficiency of gait analysis in medical and rehabilitation applications. Similarly, another study used a

markerless motion capture system based on deep learning algorithms to simultaneously record the gait of thirty healthy adult participants on a treadmill with eight video cameras and seven infrared optical motion capture cameras. The measured gait kinematics were compared with data obtained from a standard marker-based motion capture system, showing similarity in lower extremity joint angles. Markerless data acquisition methods were deemed a practical alternative [23]. Another AI-specific application by Needham et al. [24] developed a multi-camera, markerless fully automatic, deep learning-based motion capture system. They compared their system with a marker-based OpenSim analysis system. Their comparisons of running, walking, and jumping motion patterns recorded an average angle difference of 0.1–10.5 degrees for the hip joint and 0.7–3.9 degrees for the knee and ankle joints.

OpenPose emerges as an innovative open-source software for markerless motion capture. With its modular structure and advanced computer vision capabilities, OpenPose offers real-time 2D pose detection and can identify numerous key points on the human body [25]. The software continuously improves, with recent updates significantly enhancing accuracy [26]. By incorporating new key points defined for various body movements, OpenPose provides an accessible, precise, and flexible markerless motion capture solution. Its open-source nature allows for ongoing development, supporting advancements in biomechanics and computer vision [27].

In a study by Haberkamp et al. [28], OpenPose was used to determine joint angles during a single-leg squat without markers, using 2D and 3D images. The results showed moderate to strong correlations between 2D pose estimation and 3D techniques for sagittal plane joint angles. Similarly, Hsiao et al. [29] used OpenPose to analyze joint kinematics during a push-up motion with a single camera, without markers. They evaluated elbow joint angles and used fuzzy logic to classify joint movements. This study highlights potential applications in assessing rehabilitation exercises and industrial safety, identifying inappropriate or hazardous movements in real-time.

Hernandez et al. [30] compared the accuracy of OpenPose with Detectron 2, evaluating these low-cost, markerless human pose estimation tools. The study aimed to assess elbow and shoulder joint angles predicted by OpenPose and Detectron 2 during four different upper extremity rehabilitation exercises. Using two Kinect 2 RGBD cameras, the experiments showed that OpenPose performed better than Detectron 2.

The features and results of some similar studies on motion capture methods for human movement analysis are presented in Table 1. Evaluating these studies reveals that deep learning-based motion estimation techniques offer significant potential for real-time, markerless motion analysis. When reviewing the studies summarized in Table 1, it can be seen that many of the motion capture methods developed have certain advantages and disadvantages. Marker-based and IMU-based systems offer high accuracy but come with practical challenges such as complex setup, calibration, and the need for specialized personnel. While depth camera systems like Kinect offer more accessible setups, they have relatively high costs and environmental limitations such as ambient lighting and detection range. Deep learning-based markerless approaches—particularly OpenPose and Theia3D—show promising results in terms of flexibility and ease of use, but they still face challenges related to high computational power and environmental robustness. Additionally, these studies do not include joint moment calculations by directly integrating pose estimation with biomechanical modeling.

Table 1 Motion capture methods for human movement analysis

Ref	Movement	Equipment	Algorithm-Method	Performance
[31]	Walking	Kinect V2 + IMU	Kinect SDK Mathematical Computation	RMSE < 5.5°(± 2.3) RMSE < 5.6°(± 2.0)
[32]	Runing	IMU	CNN	RMSE < 3.5°
[33]	Walking	Kinect V2	Kalman Filter	RMSE ≈ 4.0°
[34]	Walking	RGB Camera	Kinovea + VICON	ICC > 0.9
[35]	Walking	IMU	Sensor fission + Softening + Filtering	RMSE < 2.56°
[36]	Walking, Runing, Kneeling down	IMU	Xsense, Visual3D, MATLAB	Lower ext. RMSE < 9°, Upper ext. RMSE < 15°
[37]	Single Leg Kneeling	IMU	OptiTrack, iOS, SportsLapp	r = 0.929–0.988, p < 0.001
[38]	Walking	RGB Camera	OpenPose, Kinovea, MAC3D	Hip and knee ICC > 0.89
[23]	Upper extremity rehabilitation exercise	Kinect V2 (x2)	OpenPose, Dedectron 2	OpenPose RMSE < 7.9° OpenPose RMSE < 14.18°
[30]	Walking	RGB Camera (x8)	Theia3D, MATLAB	RMSE < 5.5°
[39]	Walking, Runing, Cycling	Virtual RGB Camera (x4)	OpenPose, OpenSim	RMSE < 1.6°
[40]	Load lifting	RGB Camera	VideoPose3D	r > 0.9, RMSE < 10°
[41]	Knee exercise	Wearable textile-based sensor	MATLAB	RMSE < 2.0°
[20]	Walking	Wearable IMU	OpenSenseRT	Lower ext. RMSE < 4.4° Upper ext. RMSE < 5.6°
[42]	Gait	Single camera	OpenPose, LSTM	Accuracy 95%

A general review of the studies in the literature reveals that various methods for motion capture and joint moment analysis exist, each with its advantages and limitations. Marker-based systems provide high accuracy in joint kinematics but are limited in terms of accessibility and practicality in daily clinical or sports environments due to the need for specialized equipment and complex setup. Markerless methods, on the other hand, make movement recording more natural, reducing setup time and costs, but may encounter issues such as lighting conditions, background complexity, and joint point detection. These systems may exhibit lower accuracy in dynamic movements. Furthermore, although high correla-

tions have been shown between marker-based and markerless systems, both methods still require further improvement in calculating joint moments. One common limitation of many systems is the difficulty in performing real-time analysis in clinical settings. Markerless systems need to be made more robust under various conditions, and improvements are necessary in areas such as integrating real-time feedback into clinical decision support systems and reducing setup complexity. Markerless motion capture systems require a rapid setup for detecting human movement. Additionally, it is crucial for these systems not only to detect joint points but also to determine joint moments, which are an important indicator for biomechanical analysis.

Table 1 shows that OpenPose is frequently chosen for markerless motion analysis in previous studies. While OpenPose provides valuable results in motion analysis, the MediaPipe algorithm stands out with several notable features. MediaPipe was chosen over OpenPose due to its ease of integration, computational efficiency, and suitability for real-time applications. MediaPipe provides a seamless pipeline for pose estimation with built-in tracking and keypoint detection, making it highly compatible with various platforms and devices. Its lightweight architecture allows faster processing speeds even on standard hardware, reducing the need for high-end GPUs. In contrast, while OpenPose is highly accurate and customizable, it typically requires more computational power and involves a more complex setup process. MediaPipe also supports direct implementation through common programming environments like Python, simplifying the development process. Given our focus on creating a cost-effective and easily deployable system for STS analysis, MediaPipe offers a practical balance between accuracy and usability [43–45].

Considering the above, one of the fundamental requirements in motion analysis is the accurate estimation of joint angles and moments without using expensive and restrictive marker-based systems. Traditional marker-based systems such as Vicon are widely used due to their high accuracy; however, these systems require complex setups and specialized personnel. Additionally, markerless approaches have been developed that incorporate depth cameras (e.g., Kinect), IMUs, and deep learning-based applications such as OpenPose or MediaPipe. While OpenPose demonstrates strong performance in 2D pose estimation, it typically requires high processing power and multi-camera systems. MediaPipe, on the other hand, offers a real-time, low-resource solution using a single RGB camera. However, most MediaPipe applications lack integrated joint moment calculations with biomechanical models.

Previous studies have demonstrated the potential of markerless applications such as OpenPose and Kinect-based systems in estimating joint kinematics, but these applications have some limitations. These include high computational requirements, the need for multiple cameras or sensors, sensitivity to lighting and background conditions, and the lack of integration with dynamic biomechanical models for joint moment calculation. Additionally, most of these systems require complex setups or are not suitable for real-time clinical environments. Our work addresses these drawbacks by proposing a low-cost, single-camera system that not only estimates joint angles but also calculates joint moments by integrating with an inverse dynamics model using MediaPipe. This offers a more accessible approach to human motion analysis, particularly in scenarios such as STS movements, which are commonly used in rehabilitation assessments.

This study introduces a novel application of MediaPipe, a deep learning-based library, for the markerless estimation of joint angles and the subsequent calculation of joint moments

during STS movements. Unlike conventional motion capture systems, which necessitate the use of cumbersome markers and costly equipment, our approach employs a single, readily available camera to capture high-resolution video footage. This footage is then processed to extract precise joint kinematics. This methodology not only streamlines the setup process and reduces financial burdens but also enhances the user experience by removing the movement constraints typically imposed by markers. By integrating MediaPipe's advanced pose estimation capabilities with a comprehensive inverse dynamics model, this research bridges the gap between sophisticated biomechanical analysis and practical, real-world applications. This research has the potential to simplify motion analysis and make it more accessible to a broader range of users, including physiotherapists, sports scientists, and clinicians. The proposed system in this study enables biomechanical analyses with minimal technical details.

Lastly, by detailing the application we developed and sharing our experiences, we evaluate the effectiveness and accuracy of our application, which uses a single camera to estimate angles from images, calculate angular velocity and acceleration, and determine joint moments based on an inverse dynamics model. This study aims to contribute to the development of more cost-effective and accessible methods in motion analysis and provide significant insights into the analysis of the sit-to-stand movement, ultimately assisting physiotherapists and sports scientists in their decision-making processes. The main contributions of this study to the literature are as follows:

- A low-cost, markerless motion analysis system was developed using a single RGB camera and the MediaPipe framework for joint angle estimation during STS movements.
- An inverse dynamics model was integrated with MediaPipe outputs to compute joint moments at the hip, knee, and ankle joints.
- The accuracy of the proposed method was validated by comparing it with a marker-based system and MATLAB Multibody simulations.
- A detailed performance evaluation was conducted under varying video resolutions, lighting conditions, and hardware configurations to demonstrate the robustness and applicability of the system in clinical and rehabilitation settings.

2 Materials and methods

2.1 Motion analysis of the STS

The sit-to-stand (STS) movement is frequently used in human movement analysis studies. Our experiment included 15 healthy volunteer subjects. Among the subjects, 12 were male and 3 were female. The ages of the subjects ranged from 18 to 42 years, with an age average of 27.66 (±10.53) years. The height average for male subjects was 1.77 (±0.08) meters, and their mass average was 79.33 (±14.65) kilograms. The height average for female subjects was 1.60 (±0.08) meters, and their mass average was 62.66 (±7.76) kilograms.

The experiment was conducted in a consistent environment throughout. The imaging device used was the camera of a Redmi Note 9 Pro smartphone. Video recording was done at 1080p resolution, 30 fps, and in mp4 format. Previous studies have shown that 30 fps is sufficient for detecting the kinematic changes of joints during the STS movement using markerless methods [46]. The camera was placed 4 m away from the subject and positioned

75 cm above the ground. The sit-to-stand movement performed by the subjects consisted of stages from a sitting position to a fully standing position and back to sitting. The timing of the subjects' movements during the sit-to-stand and return-to-sit phases was not interfered with. Subjects were asked to keep their feet fixed during the movement. The feet were positioned parallel to each other in a symmetrical stance, and filming was done from the subject's right sagittal plane. The stages from the beginning to the end of a person's movement are shown in Fig. 1. A 60 cm high chair was used for the sit-to-stand movement. The subject performing the sit-to-stand movement was instructed to stand up and sit down without leaning back and with their arms crossed over their shoulders for support.

2.2 Pose estimation with MediaPipe and determination of joint angles

MediaPipe pose estimation was performed on hardware equipped with an Intel i5-4690 processor, 8GB DDR4 RAM, and an NVIDIA GeForce GTX 750 Ti graphics card running Windows operating system. Python [47] programming language was used for implementing pose estimation using MediaPipe. Anaconda3 [48] was utilized for managing modules and related packages. Jupyter Notebook [49] served as the editor for development. OpenCV library was employed for computer vision functionalities. NumPy [50] and pandas [51] libraries were used for matrix operations and computations, respectively.

For each participant, both markerless (Fig. 2a) and marker-based (Fig. 2b) videos were recorded. These videos were resized to 720p resolution and saved in separate folders for each participant in mp4 video format. Kinematic data was obtained separately using Kinovea 0.9.5 [52] and software developed using the MediaPipe library. As seen in Fig. 2, detecting a few key joint points during the STS movement is sufficient for joint moment calculation. Positioning a camera in the sagittal plane allows for detecting the fingertip, heel,

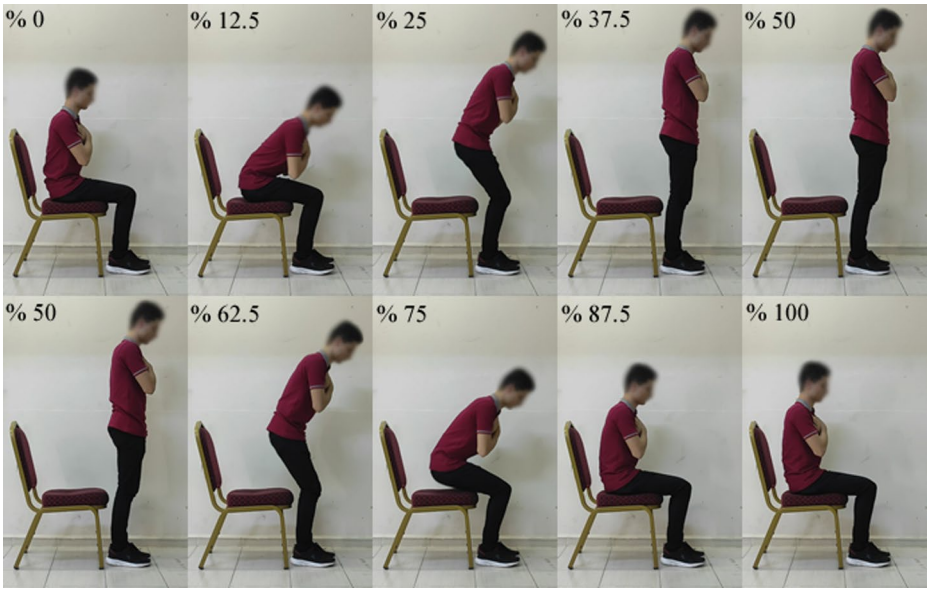


Fig. 1 Phases of the STS movement

Fig. 2 STS motion capture (**a**) markerless (**b**) marker-based

ankle, knee, hip, and shoulder joints, which is adequate for evaluating the lower extremity biomechanics during the STS task [46].

In the Mediapipe Pose library, previously annotated videos for joint detection in the Kinovea program were used. Each frame of the video was processed to obtain a sequence of three-dimensional normalized joint locations for each point. The obtained positional data was then passed through an angle calculation function to compute angle values for each of the three joint points. The calculated angle values were annotated on the visual using OpenCV codes, similar to Fig. 3. These angle values were saved into CSV files for each participant.

2.3 Calculation of the joint moments

After detecting the limb angle for ankle, knee, and hip joints; angular velocity ω as calculated in Eq. 1, where θ represents the angle between two limbs, $\Delta\theta$ is the angle difference, t denotes time, and Δt is the time difference. Here, n indicates the index of each video frame.

$$\omega_n = \frac{\theta_{n+1} - \theta_{n-1}}{2\Delta t} \tag{1}$$

After angular velocity, angular acceleration α_n for each frame with index n is calculated as in Eq. 2.

$$\alpha_n = \frac{\omega_{n+1} - \omega_{n-1}}{2\Delta t} \tag{2}$$

Fig. 3 Detection of joint points and angles with MediaPipe

A link-segment model was constructed as seen in Fig. 4 to calculate joint moments. This model assumes the body is divided into two equal and symmetric parts in the sagittal plane. Since the motion analysis is focused on the hip, knee, and ankle joints, the torso, arms, hands, and head are not included in the model. Only up to the center of mass of the body is considered to calculate the hip angle. Assumptions include constant limb lengths during motion, limbs treated as point masses located at their respective centers of mass, constant positions of mass centers, single-axis joints, and unchanged joint positions.

In Fig. 4, θ_h, θ_k, θ_a represent the hip angle, knee angle, and ankle angle, respectively. L_t, L_l, L_f denote the lengths of the thigh, leg, and foot segments. C_t, C_l, C_f indicate the distances between the proximal ends of the thigh, leg, and foot segments to their respective centers of mass. W_t, W_l, W_f represent the gravitational forces acting on the thigh, leg, and foot segments due to their masses. I_t, I_l, I_f denote the moments of inertia of the thigh, leg, and foot segments. M_h, M_k, M_a represent the torques generated at the hip, knee, and ankle joints, respectively. Anthropometric data has been determined to calculate joint moments according to the created model.

Segment height The lengths of the thigh, leg, and foot segments were calculated according to the study [53] (see Appendix Table 5).

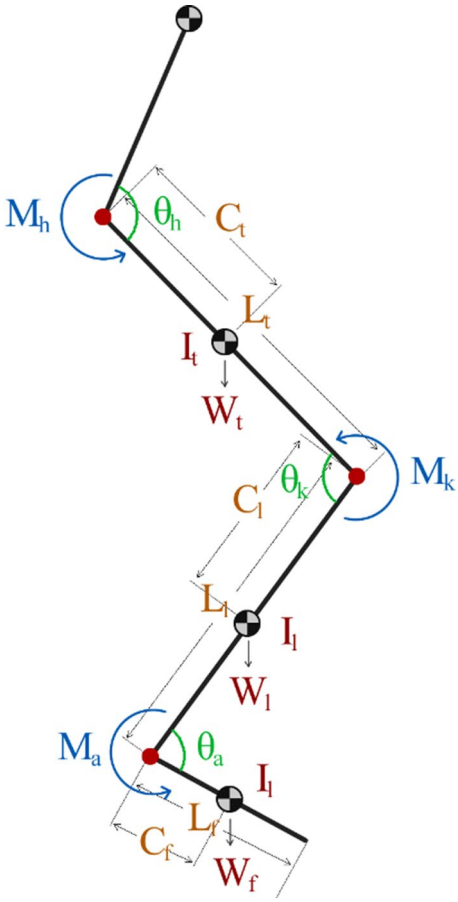


Fig. 4 The link-segment model for dynamic analysis of STS

Segment mass The masses of the thigh, leg, and foot segments were calculated according to the study [53] (see Appendix Table 6).

Center of mass The parallel axis theorem [53] was used to determine the centers of mass relative to the proximal ends of the segments (see Appendix Table 7).

Moment of inertia The parallel axis theorem [53] was used to calculate the moments of inertia of the segments about their joint rotation centers (see Appendix Table 8).

Most body segments rotate not around their centers of mass but around joint points at both ends. Measurements of moments of inertia for human body segments are typically conducted relative to a single joint center. In this context, the moments of inertia of segments (I) are calculated using the parallel axis theorem, based on the moments of inertia around the segments' centers of mass (I_0). In this study, the centers of mass and moments of inertia of the segments were also determined according to this theorem. In this theorem, the moment of inertia of the segments I is calculated as $I = I_0 + mx^2$, where m is mass of segment, x is distance between center of mass and center of rotation.

Using the parameters listed above and the joint angle data obtained from the motion analysis, joint moments were calculated according to the principle of conservation of momentum (Eq. 3). The calculation of joint moments was based on modeling limb segments as rigid bodies with fixed centers of mass and constant moments of inertia, in accordance with standard biomechanical studies. Although human limbs exhibit elastic properties due to soft tissue movement, rigid body mechanics methods are commonly applied in biomechanical analyses. This simplification allows the implementation of well-established inverse dynamics models, enabling successful estimation of joint moments. While this approach disregards dynamic changes in mass distribution during movement, its effectiveness in estimating joint moments for tasks like STS has been supported by previous research [54, 55].

$$M = I \bullet \alpha + m \bullet g \bullet cos\theta \bullet C \tag{3}$$

When Eq. 3 is adapted sequentially for hip, knee, and ankle joint moments, calculations were performed using the following Eqs. (4–6). Here, subscripted letters *h* represents the hip joint, *k* represents the knee joint, and *a* represents the ankle joint. *t*, *l*, and *f* correspond to the thigh, leg, and foot segments, respectively. In Eqs. 4–6, the values for *m* were calculated based on Appendix Table 6, *C* from Appendix Table 7, and *I* from Appendix Table 8. Additionally, the limb lengths used in the calculation of the moment of inertia were obtained from the segment heights in Appendix Table 5.

$$M_h = I_t \bullet a_{n,t} + m_t \bullet g \bullet cos\theta_{n,t} \bullet C_t \tag{4}$$

$$M_k = I_l \bullet a_{n,l} + m_l \bullet g \bullet cos\theta_{n,l} \bullet C_l \tag{5}$$

$$M_a = I_f \bullet a_{n,f} + m_f \bullet g \bullet cos\theta_{n,f} \bullet C_f \tag{6}$$

In order to compare joint moment predictions, the simulation process of STS was also conducted using MATLAB Multibody tools. Although MATLAB Multibody Simscape is primarily designed for the analysis of mechanical systems, it is also successfully applied in biomechanical analyses of human movements [56, 57]. The link-segment model (see Fig. 4) was constructed using MATLAB block diagrams as depicted in Fig. 5. The model consists of 3 segments (foot, leg, and thigh) and 3 joints (toe tip, ankle, knee). Specifically, the ankle and knee joints were modeled as single degree-of-freedom revolute joints. The toe tip was represented as a fixed joint. Based on the motion analysis results, the displacements of the knee and ankle joints were used to simulate the MATLAB model. From the simulation, joint moments during STS were computed for the knee and ankle joints. The joint moments calculated from this simulation were compared with the joint moments computed using Eqs. 4–6. In both calculations, the angular variations of joint points detected by the MediaPipe model were used. This approach was chosen to evaluate the accuracy of different computational methods and to examine the usability of the MediaPipe-based markerless system in biomechanical analyses.

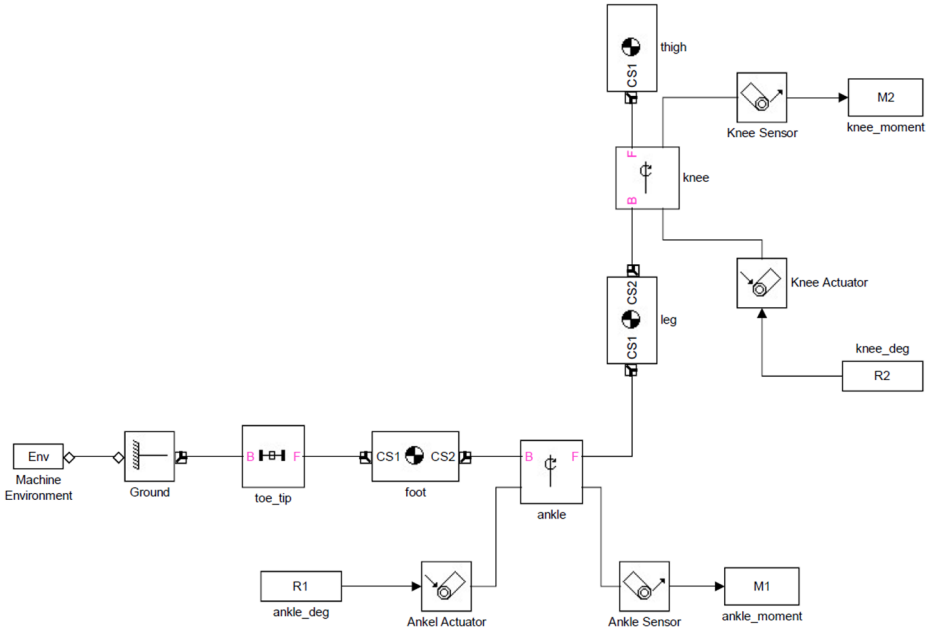


Fig. 5 MATLAB Multibody Simscape model of STS

3 Results and discussion

In this section, first, the prediction speeds of STS motion analysis using the MediaPipe library on different hardware configurations are presented. Subsequently, results obtained from videos of varying resolutions are compared. The comparison includes joint angles, angular accelerations, and moment values extracted from videos captured under different lighting conditions. Next, a detailed comparison is made between the markerless predictions using MediaPipe and the motion analysis computed using Kinovea. Finally, a comparison is conducted between joint moments calculated both according to the link-segment model and through MATLAB simulation.

3.1 Detection of joint angles using different hardware and conditions

The speed of the motion calculation application was initially tested in the measurements. The time taken for MediaPipe library to detect joint points was measured. Measurements were conducted separately for each frame to detect points and recorded. Videos were tested at three different resolutions: 1080p, 720p, and 480p. The speed test was performed on two different devices: Device1: i5 4690 3.5 GHz processor, 8 GB RAM, Windows 10 operating system. Device2: i5 8250U 1.6 GHz processor, 12 GB RAM, Windows 11 operating system. Observations were made for Device2 both while on battery power and plugged in. The speed of calculation for each frame on Device1 was determined in frames per second (fps). Additionally, video footage was captured in both low-light and well-lit conditions to observe prediction speeds. The minimum (min), maximum (max), and average (avg) fps values for each frame detection are presented in Table 2.

Table 2 Minimum, maximum, and average Fps values for device 1, device 2 plugged in, and device 2 on battery, across 1080p, 720p, and 480p resolutions, and bright, normal, and dark lighting conditions

	Device1			Device2 plugged in			Device2 on battery		
Video	min (fps)	max (fps)	avg (fps)	min (fps)	max (fps)	avg (fps)	min (fps)	max (fps)	avg (fps)
1080p Normal	3.09	15.62	12.62	3.07	18.08	15.62	1.10	12.19	8.20
1080p Bright	3.90	15.38	12.72	3.60	19.84	15.35	2.14	11.86	8.61
1080p Dark	3.87	15.86	12.80	3.91	18.65	15.67	2.49	13.78	8.87
720p Normal	3.48	16.95	15.83	3.18	19.63	15.90	2.08	13.61	9.52
720p Bright	3.64	16.66	15.82	3.75	20.63	15.83	1.7	13.47	8.72
720p Dark	3.83	16.66	15.78	3.54	19.65	15.82	1.47	15.02	8.95
480p Normal	3.58	16.66	15.74	3.35	22.25	18.78	2.52	15.79	9.80
480p Bright	3.67	16.39	15.70	3.19	22.17	19.28	1.90	16.17	10.26
480p Dark	3.74	16.66	15.78	3.54	22.00	18.92	2.15	16.20	10.16

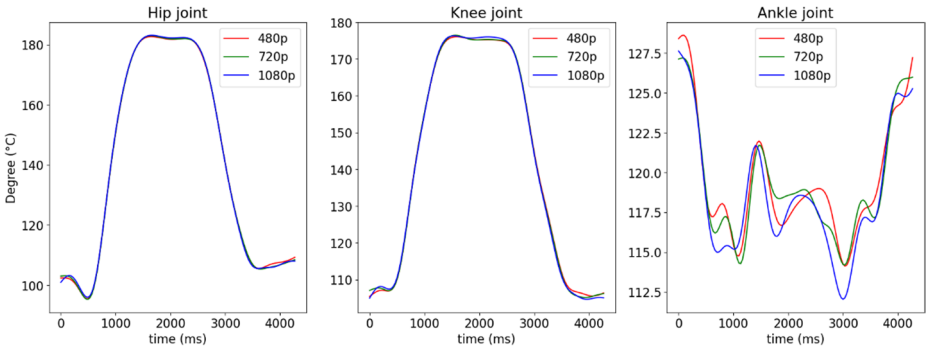


Fig. 6 Hip, knee and ankle angles obtained with MediaPipe from 480p, 720p and 1080p videos with respect to time

When initial predictions of joint points began, Device 1 and Device 2 achieved approximately 3.5 fps for all video resolutions and under different lighting conditions while operating on full power. When Device 2 operated on battery, this rate dropped to as low as 1.1 fps. Minimum fps was observed in the first frame for all devices. This is because the MediaPipe algorithm initially detects joint points in the first frame and then searches around those points in subsequent frames, reducing the detection time in subsequent frames. Different lighting conditions were observed not to significantly affect prediction time. While there was no noticeable difference in speed between 720p and 480p resolutions, the prediction speed decreased at 1080p resolution. It was observed that the power status of the device significantly affected the prediction time.

480p, 720p, and 1080p resolution videos were used to estimate joint angles, and the results are observed in Fig. 6. The estimated angle values for the hip and knee joints show similarity to each other. However, there is noticeable variation in ankle joint angles. The hip joint ranges between 95 and 180 degrees, the knee joint between 105 and 175 degrees, and the ankle joint between 112 and 128 degrees. When examining the relative angular displacements of the joints, the largest displacements were observed in the hip and knee joints, while the smallest displacement occurred in the ankle joint. These findings are consistent

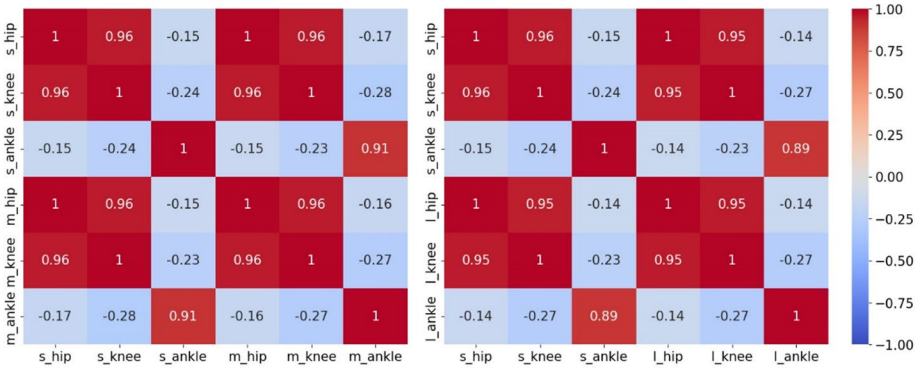
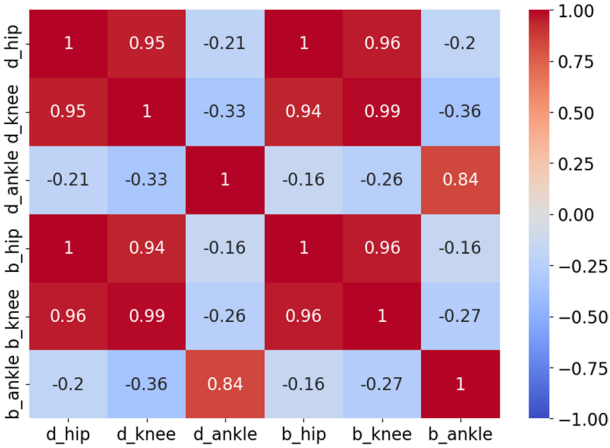


Fig. 7 The relationship between the joint moments of the videos in different resolutions

Fig. 8 The relationship between joint moments in bright and dark images



with the movement pattern of the STS task [58]. Additionally, in our study, the joint angles calculated using the MediaPipe algorithm were found to be compatible with those obtained from studies employing the OpenPose algorithm [59]. Moreover, the joint movement analysis results obtained using a single camera were also consistent with motion analysis results conducted with other hardware systems [15].

It is observed that the moments of the hip joint, knee joint, and ankle joint calculated from videos of different resolutions (480p, 720p, and 1080p) exhibit high consistency with each other. (Fig. 7).

In 720p resolution, when joint angles are estimated from normal and dark videos, there is a significant overlap between knee and hip joint angles, whereas ankle joint angles exhibit differentiation. This distinction is also reflected in the calculation of joint moments. The computed moment values for the hip and knee joints are highly consistent with each other. However, variability is observed in the ankle joint. The relationship between joint moments for both lighting conditions is depicted in Fig. 8. The correlation coefficient for hip joints is determined to be a positive 1. For knee joints, a high value of 0.99 is obtained. In contrast, ankle joints exhibit a lower correlation coefficient of 0.84, which, while acceptable,

is notably lower compared to the other joints. In this context, lighting conditions and complex backgrounds can cause estimation errors in markerless pose estimation systems like MediaPipe. Low lighting may create shadows or low-contrast areas, making joint detection more challenging. Similarly, a visually cluttered background can result in incorrect keypoint associations. These factors can particularly affect distal joints such as the ankle, where precise detection of subtle positional changes is critical. Implementing environment-specific adjustments or employing more robust noise-reduction algorithms could further enhance the system's reliability under varying conditions. For example, the TomFusioNet al.gorithm proposed by Kaushik et al. [60] can be used to reduce background noise. Here, a background subtraction technique based on the hue, saturation, and value (HSV) color model is suggested according to the importance of relevance.

In addition, pose estimation from videos, especially at low frame rates like 30 fps, can sometimes result in erroneous transitions between joint points instead of smooth transitions. This can lead to errors or inconsistencies in joint angle predictions, especially during fast movements or transitions. However, for the relatively slow and controlled STS movement, the margin of error introduced by a 30-fps frame rate is minimal and does not affect joint angle and moment calculations. Future studies may prefer to use higher frame rates to improve prediction accuracy and reduce error margins in high-speed activities.

Although the proposed system was designed with real-time applicability in mind, no formal latency or throughput measurements were performed in this study. The computational efficiency of MediaPipe and the inverse dynamics implementation suggests feasibility for real-time operation on modern hardware. However, actual deployment on embedded systems or mobile platforms would require further optimization and benchmarking.

3.2 Comparison of joint angles with and without markers

Figure 9 compares the estimated angles of hip, knee, and ankle joints obtained both through MediaPipe (markerless) and Kinovea software (with markers). It is observed that knee and hip joint angles are similarly detected by both Kinovea and MediaPipe. However, there is a discrepancy in ankle joint angle estimation. This difference arises because in marker-based motion analysis using Kinovea, the toe tip point remains fixed, whereas in markerless motion analysis with MediaPipe, this point undergoes oscillation. During the STS movement, the ankle joint can be identified more consistently in marker-based systems due to the use of physical markers placed on anatomically prominent landmarks. In contrast, MediaPipe relies on visual cues to estimate joint positions, and the anatomical prominence of the ankle is often less distinguishable due to occlusion, low contrast, or individual anatomical

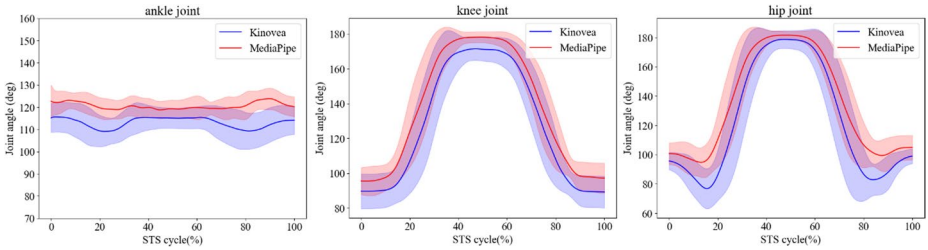


Fig. 9 Comparison of joint angles calculated with Kinovea and MediaPipe

variation. This leads to greater variability in ankle key point localization, thereby reducing the accuracy of ankle joint angle estimation.

3.3 Comparison of joint moments

Figure 10 provides a detailed visualization of the joint moments during the STS cycle for the ankle, knee, and hip joints. The top row of the figure illustrates the variability in joint moments across 15 participants. Each plot (ankle, knee, and hip) shows individual trajectories of joint moments, highlighting the significant inter-individual differences in the biomechanics of the STS movement. In the bottom row, the figure compares the average joint moments derived from Kinovea and MediaPipe. The blue lines represent the joint moments calculated using the Kinovea system, while the red lines correspond to those estimated by MediaPipe. The shaded areas around each line indicate the standard deviation, providing insights into the variability and reliability of the measurements.

In this study, we used a traditional inverse dynamics approach based on rigid body assumptions and constant inertia parameters obtained from standard biomechanical models. Although such simplifications are widely accepted in human motion analysis, they do not reflect soft tissue deformation, joint flexibility, or segmental variability seen in STS movements.

When examining the calculated moment values obtained from Kinovea and MediaPipe, a significant relationship is observed between hip and knee joint moments. However, there is a weak correlation (R) observed for ankle joint moments (Fig. 11). Upon reviewing the calculated correlations, a strong positive relationship of 0.94 is observed between hip joint moments. For knee joint moments, this relationship is also positive at 0.95. In contrast, a correlation coefficient of 0.11 is observed for ankle joints. In addition to correlation analysis, error metrics were computed. Table 3 shows performance metrics for joint moments. The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) values were lowest

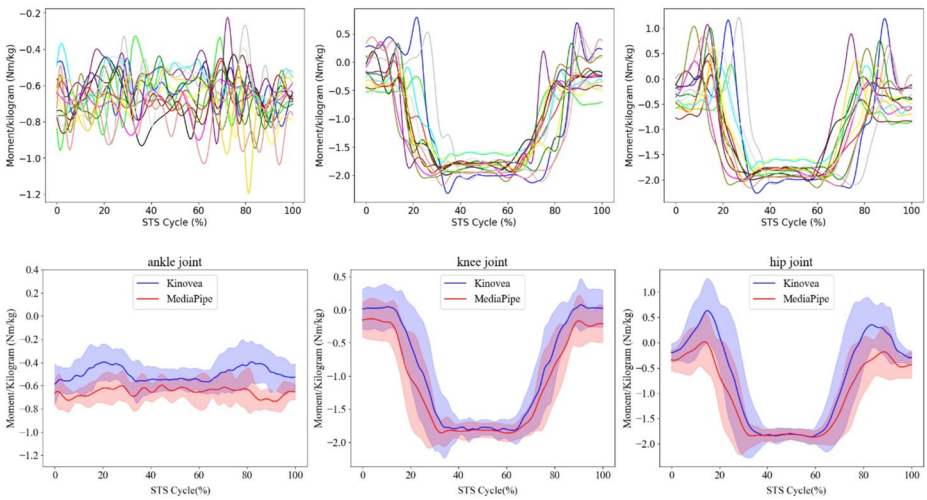


Fig. 10 Comparison of Joint Moments. Top: Joint moments calculated using the link-segment model for participants. Bottom: Mean moment values calculated based on joint angles obtained using Kinovea and MediaPipe

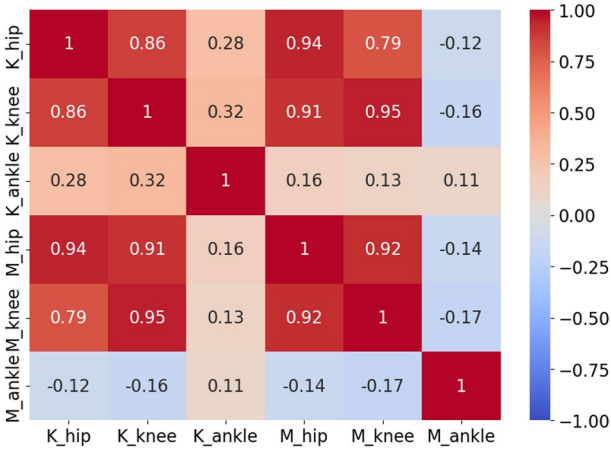


Fig. 11 Correlation coefficients of moment values for the participants

Table 3 R, RMSE and MAE comparation between markerless and marker-based joint moment Estimation

Joint	*R*	RMSE	MAE
Hip	0.94	0.49	0.34
Knee	0.95	0.35	0.24
Ankle	0.11	0.23	0.19

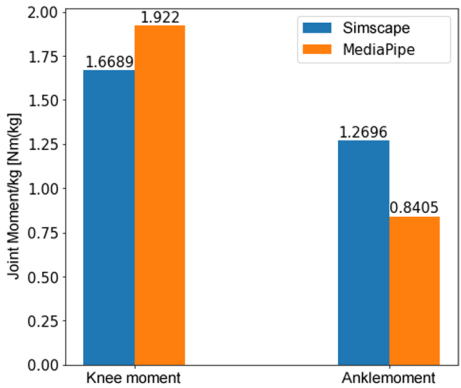


Fig. 12 Comparison of maximum moments in ankle and knee joints

for the knee and hip joints, supporting the strong correlation findings. Despite the low correlation coefficient observed for the ankle joint (R=0.11), the RMSE and MAE values for this joint were lower than those for the hip and knee joints. This difference can be attributed to the ankle joint moments being generally smaller and less variable during the sit-to-stand task. This results in smaller absolute errors.

Figure 12 illustrates the comparison of joint moments calculated using the link-segment model and MATLAB simulation results. Due to the absence of the torso segment in the MATLAB Multibody simulation, only ankle and knee joint moments could be computed. It is observed that knee joint moments calculated both with MATLAB Multibody simulation and the link-segment model are at similar levels. However, there is a significant difference in ankle joint moments. This is believed to be because the toe tip point is considered as a

fixed hinge in the MATLAB model, whereas in the MediaPipe algorithm, the position of the toe tip point varies continuously. This discrepancy leads to different calculations of ankle joint moments.

The moment values per unit weight calculated for hip, knee, and ankle joints are observed in scatter plots in Fig. 13. Ankle joint moments exhibit a normal distribution. However, hip and knee joint moments do not conform to a normal distribution. Hip moment values show a linear relationship close to positive within themselves. The same observation applies to knee moment values, where there is even a significant positive relationship between hip and knee moments.

Based on the non-normal distribution of knee and hip joint moments, Spearman's and Kendall's correlation coefficients were computed, resulting in the findings presented in Table 4. Both correlation methods indicated insignificant relationships for ankle joint angles. However, strong positive correlations were observed for hip and knee joint moments.

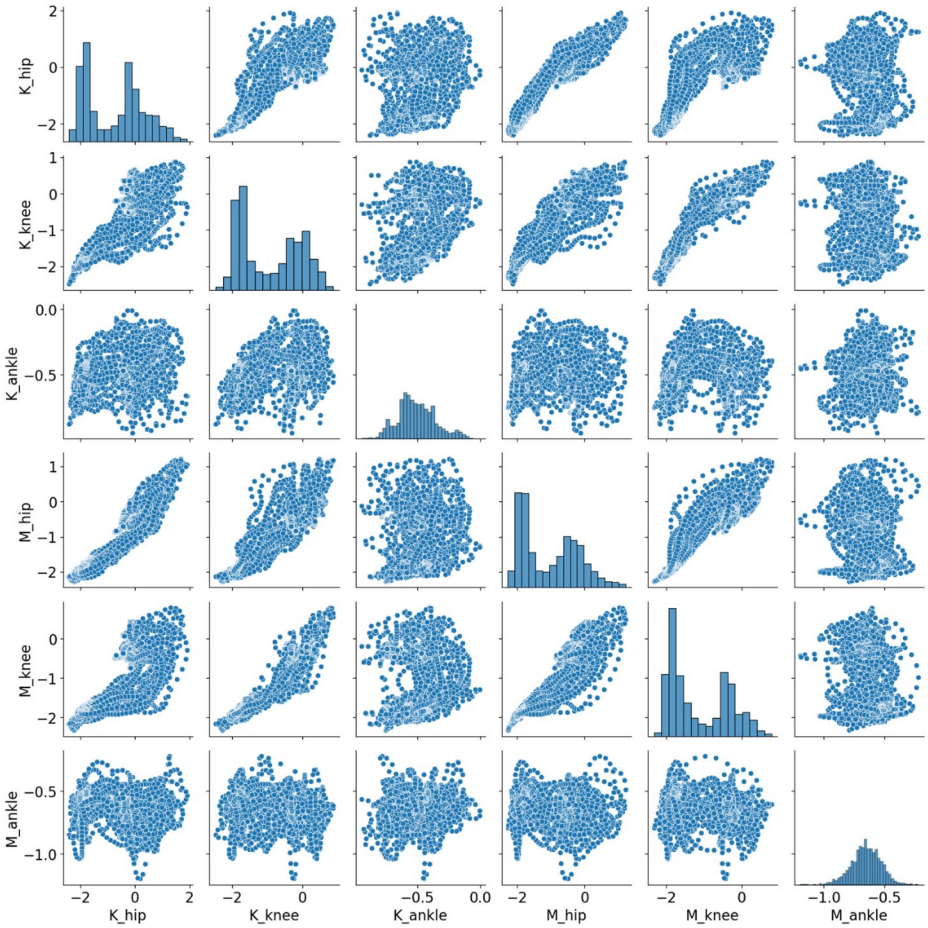


Fig. 13 Scatter plot of hip, knee, and ankle joint moments (MediaPipe)

Table 4 Spearman's and kendall's correlation coefficients for joint moments

Method	Joint	Correlation	*P* value
Spearmanr	Hip	0.932	0.0
	Knee	0.946	0.0
	Ankle	0.085	7.0
Kendalltau	Hip	0.784	0.0
	Knee	0.803	0.0
	Ankle	0.057	5.58

3.4 Clinical significance

The proposed markerless motion analysis approach has high potential for clinical application due to its portability, low cost, and minimal setup requirements. The ability to estimate joint moments using only an RGB camera makes the system applicable in various fields such as home based physical therapy, sports performance analysis, occupational health, and ergonomics assessments. Unlike traditional systems, which require expensive equipment and controlled environments, this method can be easily integrated into lightweight, user-friendly software platforms. This presents an opportunity for technology companies and healthcare providers to develop scalable and economical biomechanical analysis tools.

4 Conclusion

In this study, we demonstrated the efficacy of MediaPipe for markerless joint angle estimation and its application in calculating joint moments during STS movements. Our approach, utilizing a single camera for motion capture, proved to be a cost-effective and accessible alternative to traditional marker-based systems. The results showed strong correlations between the joint angles obtained from our markerless system and those from conventional marker-based methods, validating the accuracy and reliability of our approach.

The proposed system uses the MediaPipe infrastructure, which is designed to provide real-time performance on low- to mid-range hardware. Tests have shown speeds of over 15 fps for 720p and 480p videos, and around 8 fps in battery mode. These values are sufficient for slow and controlled STS movement. The system carries a much lower computational load compared to other methods, as it does not require a GPU or multiple cameras. MediaPipe's lightweight architecture reduces the need for recalculation by tracking joint positions in each frame. This efficient structure makes the proposed approach suitable for clinical applications with limited resources.

The integration of MediaPipe with an inverse dynamics model allowed us to accurately compute joint moments, providing valuable insights into the biomechanics of STS movements. This research highlights the potential of deep learning-based, markerless systems in various fields, including clinical biomechanics, sports science, and physical rehabilitation. The ease of setup, reduced costs, and elimination of movement restrictions associated with traditional systems make this approach particularly advantageous.

This study has some limitations. First, the MediaPipe model was applied only to STS movements, and its accuracy may vary in more dynamic or multi-planar movements. Additionally, joint moment estimates were made using a simplified inverse dynamics model that includes lower body dynamics. In future studies, the method could be extended to different

movements (walking, squatting, stair climbing, etc.) and integrated with more comprehensive biomechanical models (e.g., whole-body models). Another future work will also focus on deploying and evaluating the proposed method on mobile or embedded systems to assess its real-time performance and practical applicability in field or clinical settings.

Appendix

Table 5 Total height and segment length of the participants (m).

No	Height (H)	Thigh Length (L_t)	Leg Length (L_l)	Foot Length (L_f)
1	1.80	0.4410	0.4428	0.2736
2	1.69	0.4141	0.4157	0.2569
3	1.75	0.4288	0.4305	0.2660
4	1.92	0.4704	0.4723	0.2918
5	1.54	0.3773	0.3788	0.2341
6	1.55	0.3798	0.3813	0.2356
7	1.73	0.4239	0.4256	0.2630
8	1.83	0.4484	0.4502	0.2782
9	1.70	0.4165	0.4182	0.2584
10	1.86	0.4557	0.4576	0.2827
11	1.69	0.4141	0.4157	0.2569
12	1.74	0.4263	0.4280	0.2645
13	1.67	0.4092	0.4108	0.2538
14	1.87	0.4582	0.4600	0.2842
15	1.74	0.4263	0.4280	0.2645

Table 6 Total mass and segment mass of the participants (kg)

No	Total mass	Thigh mass (m_t)	Leg mass (m_l)	Foot mass (m_f)
1	110	11.0000	5.1150	1.5950
2	69	6.9000	3.2085	1.0005
3	85	8.5000	3.9525	1.2325
4	91	9.1000	4.2315	1.3195
5	54	5.4000	2.5110	0.7830
6	65	6.5000	3.0225	0.9425
7	77	7.7000	3.5805	1.1165
8	64	6.4000	2.9760	0.9280
9	71	7.1000	3.3015	1.0295
10	90	9.0000	4.1850	1.3050
11	56	5.6000	2.6040	0.8120
12	82	8.2000	3.8130	1.1890
13	81	8.1000	3.7665	1.1745
14	82	8.2000	3.8130	1.1890
15	63	6.3000	2.9295	0.9135

Table 7 Distance of the centers of mass (CM) relative to the proximal ends of the segments (m)

No	Height (H)	Thigh CM (C_t)	Leg CM (C_l)	Foot CM (C_f)
1	1.80	0.1910	0.1917	0.1368
2	1.69	0.1793	0.1800	0.1284
3	1.75	0.1856	0.1864	0.1330
4	1.92	0.2037	0.2045	0.1459
5	1.54	0.1634	0.1640	0.1170
6	1.55	0.1644	0.1651	0.1178
7	1.73	0.1835	0.1843	0.1315
8	1.83	0.1941	0.1949	0.1391
9	1.70	0.1803	0.1811	0.1292
10	1.86	0.1973	0.1981	0.1414
11	1.69	0.1793	0.1800	0.1284
12	1.74	0.1846	0.1853	0.1322
13	1.67	0.1772	0.1779	0.1269
14	1.87	0.1984	0.1992	0.1421
15	1.74	0.1846	0.1853	0.1322

Table 8 Moments of inertia (MI) relative to the proximal end of the limbs of the participants $(kg.m^2)$

No	Thigh MI (I_t)	Leg MI (I_l)	Foot MI (I_f)
1	0.6243	0.2794	0.0568
2	0.3452	0.1546	0.0314
3	0.4559	0.2041	0.0415
4	0.5876	0.263	0.0534
5	0.2244	0.1004	0.0204
6	0.2735	0.1225	0.0249
7	0.4036	0.1807	0.0367
8	0.3753	0.1681	0.0341
9	0.3592	0.1609	0.0327
10	0.5454	0.2442	0.0496
11	0.2801	0.1254	0.0255
12	0.4349	0.1947	0.0395
13	0.3959	0.1772	0.036
14	0.5024	0.2249	0.0457
15	0.3341	0.1496	0.0304

Authors' contributions Suleyman Akturk: The development of research ideas, collection and analysis of experimental data, implementation of deep learning algorithms, interpretation of findings, manuscript writing. Fatma Betul Derdiyok: Analysis of experimental data, interpretation offindings, manuscript writing. Kasim Serbest: Corresponding of the study, the development of researchideas, analysis of experimental data, interpretation of findings, manuscriptwriting, and general revision.

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Data availability Data will be made available on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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